

Supplementary Material

Arrows in Their Backs: Vocabulary Lead and Product Survival in the US Trademark Record

Eric Silver

September 2026

This supplement carries detail behind the five-year-proof results (§S1), the waves (§S2), and the capital-markets ladder (§S3). Section, table and figure numbers without an S prefix point into the article. The construction of the measure and its robustness program are documented in the companion paper and its supplement; every class’s profile is in the online appendix at <https://aporia.institute/tm-vocabulary/online-appendix/>.

S1 The five-year proof: timing, ties, the renewal, and per-class extremes

The five-year proof is close to a fixed appointment rather than a hazard spread over years: 97% of the cancellations that fall in the 4.0–8.5 year window post between registration age 6.5 and 7.0. Since the cancellation record ends on 2 April 2026, a cohort is fully observable only once it has reached seven years, which makes 2018 the last complete cohort. On the 3.15 million registrations whose ninth anniversary is inside the record — for which failure is settled and no window is needed — the contrast is +2.71pp within class and registration year, against +2.65pp on the paper’s 4.0–8.5 window; conditional on failing, the probability of failing late carries a contrast of -0.01pp ($t = -0.2$). Response speed does not predict the outcome: across the 1.22 million scored registrations with a usable office-action/response pair, passage varies between 46.1% and 48.1% across response-speed quintiles, from a median of seven days to 183, and the lead penalty is present in both halves (-3.48 ± 0.40 and $-2.36 \pm 0.40\text{pp}$ on passing, against -2.94 pooled). The restriction drops 61% of the sample, disproportionately applications that never drew a refusal.

Tied scores: within the class×registration-year cell the quintiles are cut in, 17.1% of scored registrations share a lead value with at least one other filing. Filings that share a class, a filing date and a goods/services description have identical theme mixes scored against identical references. Cut points are fixed by sorting on the score and then on the serial number; sorting ties the opposite way moves the top-versus-bottom contrast by 0.001pp, and five pseudorandom orderings move it by at most 0.002pp, against a contrast of 2.287pp.

At the year-ten renewal, fitted over 872,079 five-year survivors with a resolved status, a straight line through renewal against lead explains 1% of the variation across forty bins and a V explains 89% ($\Delta\text{AIC} = 83$); renewal peaks at 64.2% near $z = -0.4$ and falls to 45.5% at the most lagging bin and 50.6% at the most leading.

Per class, the largest raw penalties sit in tobacco (+26pp, the vaping era), hand tools (+10pp), processed foods (+10pp) and medical apparatus (+9pp), with negative lifts in beer and soft drinks, transport and construction; these are the extremes of a 44-class screen, sitting away from zero partly by selection. Every class’s profile is in the online appendix.

S2 Waves: the theme tour, internet-bearing filings, and surges

S2.1 The theme tour

In 1995–1999 the computer, software, information and data services theme failed at 64.9% against 47.5–59.0% in the other large themes. In 2000–2004 it held about 11% of the leading fifth against 40% of filings; the leading language had moved to video, music and games on electronic media (31% of the leading fifth against 12% of filings), and inside several themes the lagging fifth failed more than the leading one: -3.6pp in electronic media, -5.5 in retail store services, -6.9 in apparatus and control systems. From 2008 the leading fifth over-represented downloadable and online software 15% against 4%; the theme’s within-theme contrast in the 2008–2014 era is $+3.3\text{pp}$, and the computer-services theme’s own contrast is positive in every era ($+3.6$ to $+5.9\text{pp}$) and never reverses.

S2.2 Internet-bearing filings by class

The internet gap in event time (Figure S1): dating the internet’s arrival in each class as the first filing year the pattern reaches 2% of the class’s filings (1994–1995 in the technology and transport classes, 1999–2000 in apparel and chemicals, 2008 in machinery), the gap between internet-bearing registrations and the rest starts near $+7$ points, declines to about zero nine to eleven years after arrival, and settles between $+1$ and $+4$; across the eight classes observable in both phases the mean gap is $+4.4$ points in years zero to four and $+1.6$ from year ten, shrinking in six of the eight. Calendar time disguises this: transport (arrival 1995) peaks at $+16$ points in its 2000 cohort; machinery (arrival 2008) peaks at $+17$ in its 2012–2015 cohorts. Clothing is the standing exception: its internet-bearing registrations outlast the class in every cohort after 2004.

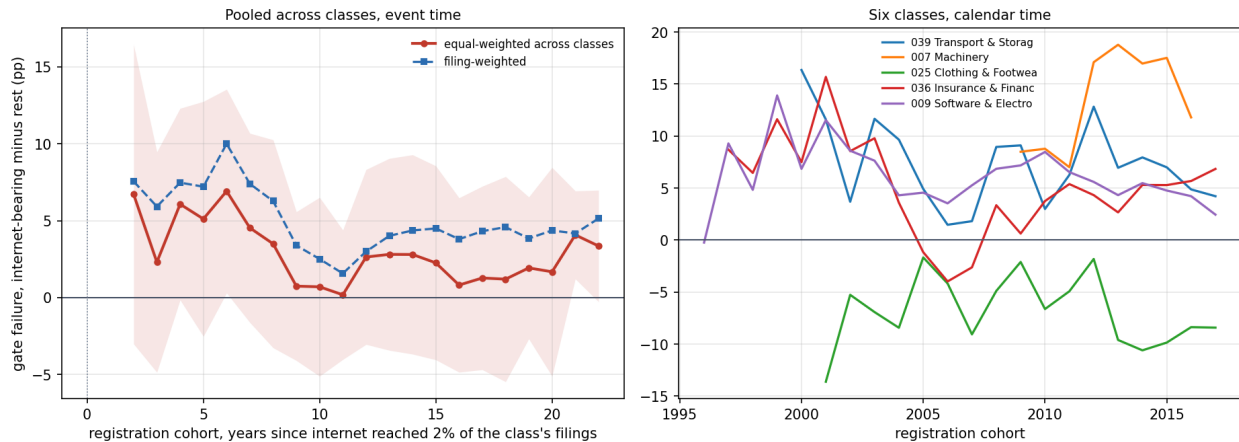


Figure S1: The internet gap in failure at the five-year proof, inside each class. *Left:* internet-bearing minus other registrations’ failure rate, by registration cohort in years since the internet pattern reached 2% of the class’s filings, averaged across classes (band: one standard deviation across classes). *Right:* five classes in calendar time. Cells with fewer than 100 registrations in either group are dropped.

S2.3 The 2000–2004 reversal, cell by cell

The cells behind Figure 8.

Table S1: Internet-bearing filings against the rest of their Nice class (Section 6.3). Registration is the share of class-records filed 1995–2018 that reached registration; failure is the share of unique registrations 2002–2018 cancelled for non-use at age 4.0–8.5 years. Classes with fewer than 500 internet-bearing registrations are omitted.

Class		Web share (%)	Registration (%)		Gate failure (%)	
			web	rest	web	rest
001	Chemicals	1.8	65.0	64.3	46.1	39.1
003	Cosmetics & Cleaning	2.8	58.0	54.2	52.1	54.4
004	Lubricants & Fuels	4.6	63.8	56.3	50.9	47.9
005	Pharmaceuticals	1.9	53.3	48.6	52.4	53.5
006	Metal Goods	3.5	68.1	67.0	44.5	39.1
007	Machinery	2.8	73.3	69.7	50.9	38.7
008	Hand Tools	3.6	65.8	62.5	42.4	43.6
009	Software & Electronics	18.7	53.3	57.0	56.5	51.4
010	Medical Apparatus	3.0	59.8	58.4	51.7	45.4
011	Lighting & Heating	3.0	67.4	64.8	51.4	45.9
012	Vehicles	3.4	69.0	60.9	49.6	45.1
014	Jewelry	7.4	63.2	55.8	54.7	55.7
016	Paper & Printed Goods	15.1	57.8	56.1	47.9	50.7
017	Rubber & Plastics	2.2	70.7	67.6	44.1	38.5
018	Leather Goods	8.0	61.4	57.4	50.5	55.4
019	Building Materials	2.0	71.4	64.0	47.5	43.9
020	Furniture	4.2	62.2	59.2	47.0	48.7
021	Household Utensils	5.1	60.1	59.0	44.5	50.5
024	Textiles	6.0	62.8	55.9	49.7	51.2
025	Clothing & Footwear	5.6	56.1	49.6	52.4	58.7
026	Lace & Embroidery	7.6	61.2	55.7	52.0	53.9
027	Carpets	5.6	66.8	56.8	43.7	46.9
028	Games & Sporting Goods	6.3	55.1	53.6	54.1	55.7
029	Meats & Processed Foods	2.5	59.7	55.7	53.3	49.6
030	Staple Foods	2.6	61.4	56.0	52.3	51.8
031	Agricultural Products	2.5	60.9	58.0	49.5	43.8
032	Beer & Soft Drinks	2.4	58.7	49.1	56.2	53.3
033	Alcoholic Beverages	1.2	61.3	52.7	56.9	42.3
034	Tobacco	2.9	53.8	49.1	55.0	54.2
035	Advertising & Retail	40.7	59.3	61.2	57.9	52.0
036	Insurance & Finance	18.9	58.6	62.6	52.4	48.9
037	Construction & Repair	10.4	67.8	67.5	49.0	47.1
038	Telecommunications	57.6	54.5	45.1	60.8	60.1
039	Transport & Storage	21.3	62.5	63.1	54.8	47.6
040	Material Treatment	9.6	68.0	63.9	49.4	47.3
041	Education & Entertainment	35.0	61.1	60.6	54.2	52.5
042	Scientific & Tech Services	39.3	57.8	59.4	58.0	51.4
043	Hotels & Restaurants	9.1	62.9	58.7	53.8	48.1
044	Medical & Beauty Services	20.6	62.9	62.1	53.7	51.3
045	Legal & Personal Services	41.5	60.3	65.0	60.1	48.7

Table S2: The 2000–2004 reversal, cell by cell. Technology-class registrations filed 2000–2004 and settled at the five-year proof; each carries its dominant $T = 50$ theme and the internet text flag of Section 6.3. Every theme×flag cell holding at least 5,000 registrations is shown. *Share* is the cell’s share of the era’s filings; *of top fifth*, its share of the era’s pooled leading fifth. Q5–Q1 is the failure contrast between the most leading and most lagging fifths, quintiles reassigned within the cell; in the bottom block, quintiles are pooled across the era (first two rows) or cut within class×cohort (last two).

Cell	n	Share	Of top fifth	Failure	Q5–Q1
services, computer, software, information (other)	62,143	24.5%	6.3%	57.3%	+2.61 (0.63)
services, computer, software, information (internet)	38,235	15.1%	5.2%	61.4%	+1.78 (0.79)
video, computer, electronic, music (other)	25,042	9.9%	27.8%	56.1%	−3.24 (0.99)
apparatus, systems, devices, electronic (other)	17,032	6.7%	2.6%	45.6%	−7.06 (1.21)
services, field, training, estate (other)	16,349	6.4%	8.9%	44.3%	+1.33 (1.23)
services, featuring, store, retail (other)	15,760	6.2%	1.5%	52.6%	−4.19 (1.25)
services, featuring, store, retail (internet)	8,365	3.3%	0.6%	54.9%	−8.25 (1.72)
design, field, systems, development (other)	7,010	2.8%	7.4%	51.8%	+4.42 (1.89)
video, computer, electronic, music (internet)	5,758	2.3%	3.3%	55.5%	−2.04 (2.08)
research, medical, scientific, testing (other)	5,560	2.2%	5.3%	51.9%	+11.78 (2.10)
All filings, pooled quintiles	253,714	100%	100%	53.4%	−1.81 (0.31)
excluding the services theme	153,336				+0.90 (0.40)
Internet-bearing, within class×cohort	63,190			58.1%	−4.52 (0.62)
Other filings, within class×cohort	190,524			51.8%	−2.87 (0.36)

S2.4 Surges

Measured against the whole corpus at a 1% share, five themes ever surge and one clears a 1,000-registration floor — online software, 1999, whose 6,339 in-window registrations failed at 63.7% (Figure S2). Within-class episodes differ in sign among themselves: the education-services surge in advertising and retail after 2002 failed 13.7 points *less* than its class; the charitable-fundraising surge in finance after 2011 failed 4.0 points more; of the episodes large enough to carry their own lead contrast, one is positive (+5.8pp, $t = 2.2$) and the rest indistinguishable from zero. On the 61 episodes with at least 300 joining registrations over a six-year entry window (64,105 of 86,447 joining registrations): joiners in the surge year fail -0.8 points relative to their wave’s mean and joiners five years in $+0.0$, with no gradient between (SEs ~ 0.5); excess failure is $+2.7$, $+3.9$ and -0.1 points across wave-size terciles, correlating with size at -0.22 ; and waves whose theme share is still at or above its surge level five years on carry $+4.3$ points of excess against $+1.0$ for receding waves.

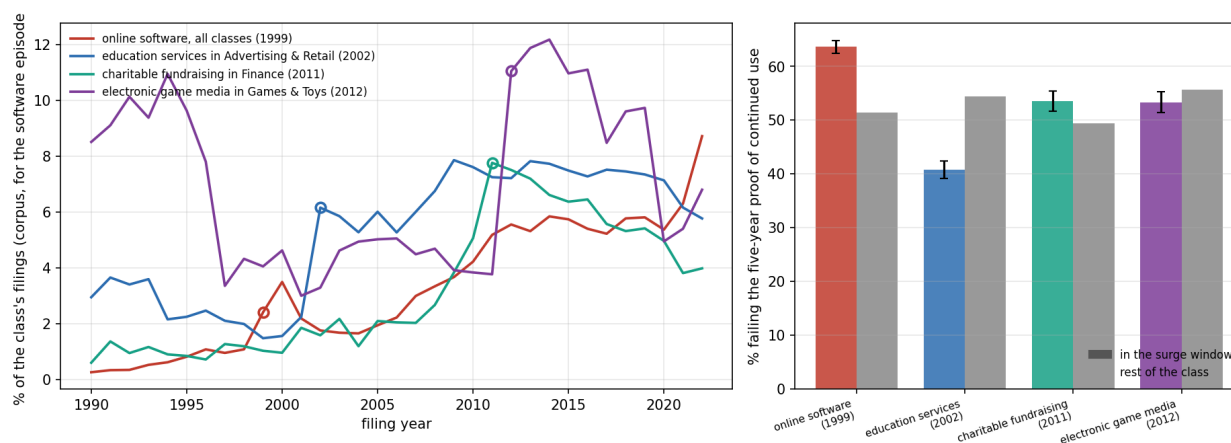


Figure S2: Four surge episodes under the 500-theme model. *Left*: the theme’s share of its class’s filings by filing year (for the online software episode, of all filings corpus-wide), on the 25% systematic sample the surge screen uses; the open circle marks the surge year. *Right*: failure at the five-year proof for registrations whose dominant theme was the surging one, filed in the three-year surge window (colored, 95% intervals), against the rest of the same class in the same cohorts (grey).

S3 Success: unfunded listings, value tiers, and curated vocabularies

The rates behind Figure 9 are in Table S3.

The coefficients behind Figure 10 are in Table S4.

Of the 1,669 debut owners with an IPO marker, 759 have no post-debut Form D round in the record; their debut lead averages -0.22 standard deviations (SE 0.04) against -0.07 (SE 0.04) for the 910 funded, 29% sit in the most lagging fifth against 22%, and both groups are mildly atypical ($+0.13$ and $+0.15$). Rounds before 2009, foreign placements, and registered offerings post no Form D.

Value tiers: ten linked owners hold 18.5% of \$19.2 trillion of linked peak revenue, fifty hold 41.5%, a hundred 53.6%, five hundred 82.9%. In standard deviations of each measure across all

Table S3: The ladder by fifth of debut lead and of debut atypicality. Rates in percent of owners whose first registered filing was made 2009–2018 ($n = 999,442$); fifths cut within Nice class, debut year, and fifth of description length. The last column is the most-minus-least contrast with its t -statistic.

Rung	Measure	Q1	Q2	Q3	Q4	Q5	Q5–Q1 (pp; t)
Funded	lead	1.74	1.61	1.50	1.46	1.69	–0.05 (–1.3)
	atypicality	1.57	1.59	1.68	1.65	1.51	–0.06 (–1.7)
Reporting	lead	0.55	0.45	0.39	0.32	0.36	–0.19 (–9.1)
	atypicality	0.39	0.34	0.41	0.43	0.51	+0.11 (+5.3)
IPO	lead	0.22	0.19	0.13	0.12	0.17	–0.06 (–4.0)
	atypicality	0.14	0.14	0.16	0.19	0.20	+0.06 (+4.9)

Table S4: Every outcome in the funnel, conditioned on its prerequisite, under one specification. Unit is the debut owner, scored on that owner’s first filing. Each row is a linear probability model with class×debut-year fixed effects and heteroskedasticity-robust errors; coefficients are percentage points per standard deviation. *Atypicality* is the average of the two KL levels; *lead* is the signed difference; the lead-magnitude column $|L|$ comes from a second specification that adds it. Cohort windows differ because each maintenance deadline needs elapsed time and Form D is electronic only from 2009, so the rows are not a nested decomposition. The five-year proof row is survival of a dated cancellation for non-use; the year-ten row is whether a registration that passed the proof was renewed rather than cancelled or expired, restricted to 2002–2013 cohorts, since a year-six and a year-ten cancellation share a terminal status and are separable only by date.

Outcome (<i>population</i>)	n	Base	Unsigned		Signed
			Atyp.	$ L $	Lead
Reached registration (<i>filed</i>)	3,161,552	56.11%	–0.982 (0.032)	–0.497 (0.047)	+0.475 (0.028)
Passed the five-year proof (<i>registered</i>)	1,260,392	41.82%	+1.311 (0.053)	–0.527 (0.083)	–0.968 (0.051)
Renewed at year ten (<i>passed the proof</i>)	297,944	55.92%	+1.383 (0.104)	–0.221 (0.156)	–0.428 (0.096)
Filed any Form D notice (<i>filed</i>)	1,535,004	2.38%	–0.125 (0.015)	+0.345 (0.027)	–0.393 (0.017)
Filed any Form D notice (<i>registered</i>)	916,448	2.45%	–0.169 (0.020)	+0.329 (0.035)	–0.324 (0.022)
Listed after the round (<i>funded</i>)	36,475	3.51%	+0.866 (0.138)	+0.734 (0.195)	–0.534 (0.129)
Became SEC-reporting (<i>registered</i>)	1,773,938	0.56%	+0.017 (0.007)	+0.023 (0.012)	–0.018 (0.007)

Standard errors in parentheses, in place of significance stars; see the note to Table 1.

debuts, with standard errors: the ten largest sit at -0.35 (0.19) on atypicality and -0.01 (0.19) on lead; the fifty largest at -0.08 (0.13) and $+0.13$ (0.16), with 28% in the most leading fifth; the hundred largest at -0.02 (0.09) and $+0.02$ (0.11), with 25% in the most leading fifth; the five hundred largest at $+0.11$ (0.05) and -0.04 (0.05), with 20% in the most leading fifth. All 4,710 linked owners average $+0.08$ on atypicality and -0.05 on lead.

Curated vocabularies: internet wording appears in 2.94 million class-filing rows, AI in 167 thousand, blockchain in 114 thousand; registration rates are 52.1%, 34.5% and 31.2% against 57.6% for all debut filings (the last two cohorts are young). On the 2009–2018 ladder, AI debuts are funded at 7.8% and blockchain at 8.0% against 2.8% for internet and 1.6% overall, while at the IPO rung the gap disappears: 0.16% of AI debuts, 0.22% of blockchain, and 0.21% of internet debuts against 0.17% overall (1,896 AI debuts, 1,366 blockchain).